

Road Accident Severity Prediction with Multimodal Data for Enhanced Road Safety

SUBMITTED BY

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1 Introduction

Road traffic accidents remain a major global challenge, inflicting not only severe human loss but also significant economic losses. With the rapid increase in vehicle density, the frequency and intensity of traffic collisions have become a major concern. Traditional approaches to traffic safety remain largely reactive, focusing on post-accident analysis rather than taking proactive measures.

Through this project we explore the application of machine learning and geospatial analysis to classify accident severity and identify high-risk, accident-prone areas using the Chicago Crash Dataset. We intend to develop a tool to equip traffic authorities and policymakers to take proactive action by integrating explainable AI tools with spatial risk analysis and an interactive web application.

1.1 Motivation

Every year, traffic accidents claim thousands of lives, imposing significant economic burdens due to injuries and property damage. Two major issues contribute to this challenge:

- **Inability to determine accident severity swiftly.** When an accident is reported, emergency teams often rely on incomplete information. Being able to estimate quickly whether an accident will result in an injury or not can significantly improve response time and resource allocation.
- **Difficulty in identifying accident-prone areas accurately.** Many cities struggle to identify which areas consistently experience severe accidents. Without data-driven insights, infrastructure upgrades may not be optimally targeted.

The motivation behind this project is to address both of these challenges using machine learning and geospatial analysis.

1.2 Novelty

Existing accident analysis systems often lack interpretability and the incorporation of spatial conditions. The novelty of this project lies in three components:

- **Combining predictive modelling with geospatial analysis.** The model not only predicts accident severity but also incorporates spatial clustering, giving insight into which localities are prone to accidents.
- **Incorporating interpretability using SHAP and LIME.** Instead of simply outputting a prediction about the severity of the crash, the system explains *why* a particular prediction was made, highlighting the influence of different features.
- **Development of a web application.** An interactive interface has been developed to obtain severity predictions from a set of input conditions.

1.3 Objectives

This project is designed with the following core objectives:

- Build a machine learning pipeline for binary accident severity classification (Injury vs. No Injury).
- Identify key risk factors and explain model predictions through SHAP and LIME analysis.
- Perform geospatial analysis to detect crash hotspots and compute risk scores.
- Develop an interactive web application for predictions and insights.

2 Project Workflow

2.1 Block Diagram of the Project

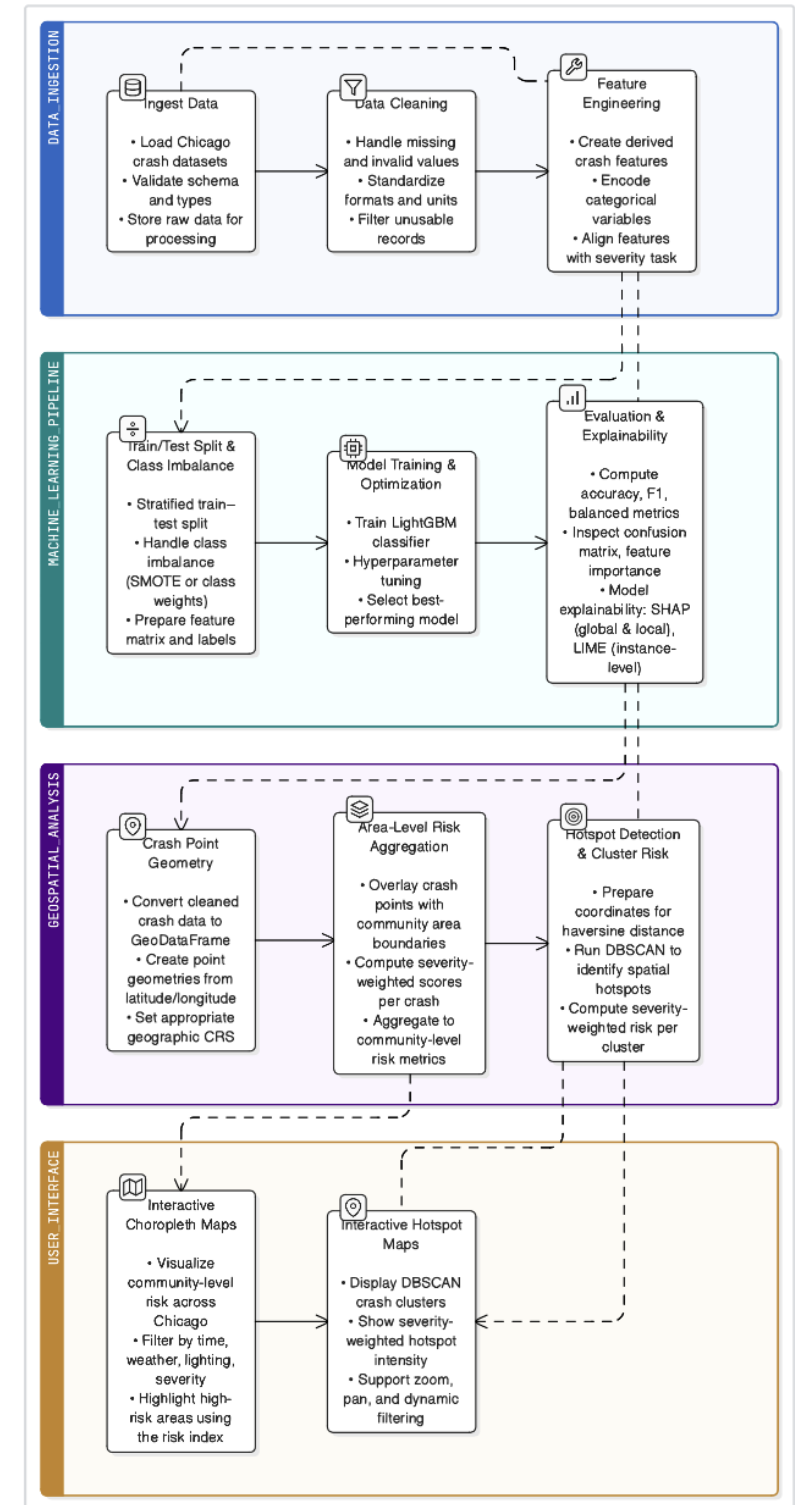


Figure 1: End-to-end project workflow: data ingestion, machine learning pipeline, geospatial analysis and user interface.

2.2 Algorithms Used in the Project

2.2.1 Modelling Algorithm

Algorithm 1 modelling.py

Input: Raw dataset D with features X and target **CRASH_TYPE**

Output: Trained best model (LightGBM), evaluation metrics, saved artefacts

1. Load the libraries required for data processing, modelling, evaluation and saving.
2. Define project directories for data, models and reports.
3. $D \leftarrow$ read the CSV file from **DATA_PATH**.
4. Assert that **CRASH_TYPE** exists in D .
5. $X \leftarrow$ all features without **CRASH_TYPE**.
6. $y \leftarrow$ **CRASH_TYPE**.
7. Save the label encoder for future inference.
8. Split (X, y) into $(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}})$ using a stratified 80/20 train-test split.
9. Define the preprocessing pipeline:
 - (a) impute missing numeric values using the **median**;
 - (b) standardise numeric features.
10. Construct a **ColumnTransformer** applying the preprocessing to all numeric features.
11. Define the model pipelines:
 - P_1 : Logistic Regression with class weights
 - P_2 : Extra Trees Classifier
 - P_3 : XGBoost Classifier
 - P_4 : LightGBM Classifier
 - P_5 : Logistic Regression with SMOTE oversampling
12. Define the evaluation metrics: **macro F1 score** and **balanced accuracy**.
13. Initialise stratified k -fold cross-validation ($k = 5$).
14. For each model $P_i \in \{P_1, \dots, P_5\}$:
 - perform cross-validation on $(X_{\text{train}}, y_{\text{train}})$;
 - compute the mean macro F1 and balanced accuracy;
 - store the results in results table R .
15. Identify the best model (**LightGBM**) from R by highest macro F1 score.
16. Fit the best model on the full training data $(X_{\text{train}}, y_{\text{train}})$.
17. Extract the top 20 most important features and save the plot.
18. Compute the evaluation metrics: test balanced accuracy, test macro F1 and the classification report.
19. Compute the confusion matrix.
20. Save the best pipeline (**joblib**), the cross-validation results (CSV) and the test metrics (JSON).
21. Return the final best model, evaluation results and saved outputs.

2.2.2 Geospatial Risk Algorithm

Algorithm 2 Geospatial risk pipeline

1. **Load the pre-processed dataset.**
2. **Create the crash GeoDataFrame:** `geometry = Point(lon, lat)`, `CRS = EPSG:4326`.
3. **Load the community boundary shapefile:** reproject to EPSG:4326 and standardise `community_area_number`.
4. **Spatial join:** join crashes to communities with `sjoin(POINT within POLYGON)`.
5. **Run DBSCAN clustering on the crash coordinates:** obtain the `cluster_id` for each crash and compute the centroid of each cluster.
6. **Compute the per-crash severity:**

$$\text{weighted_severity} = \frac{3f + 2s + 1m + 0.5n}{\text{total_injuries}}$$

where f , s , m and n are the fatal, serious, moderate and minor injury counts.

7. **Aggregate by community:** `total_crashes`, `injury_sums` and `mean(weighted_severity)`; compute the `community_risk_index` with the same weighting.
8. **Aggregate by cluster:** the same `injury_sums` and `total_crashes`; compute the cluster `weighted_score` using the risk-index formula.
9. **Visualise:** colour-code the map by community risk, and render cluster hotspot maps and the interactive Dash dashboard.

3 Experimental Results

3.1 Cross-Validation Model Performance Comparison

Table 1: Stratified 5-fold cross-validation results (reports/cv_results.csv).

Model	Macro F1	F1 std	Bal. accuracy	Bal. acc. std
Logistic Regression	0.7413	0.0018	0.7847	0.0017
Extra Trees	0.7813	0.0011	0.7702	0.0013
XGBoost	0.8037	0.0019	0.7959	0.0023
LightGBM	0.8040	0.0018	0.7966	0.0020
LogReg + SMOTE	0.7425	0.0020	0.7850	0.0019

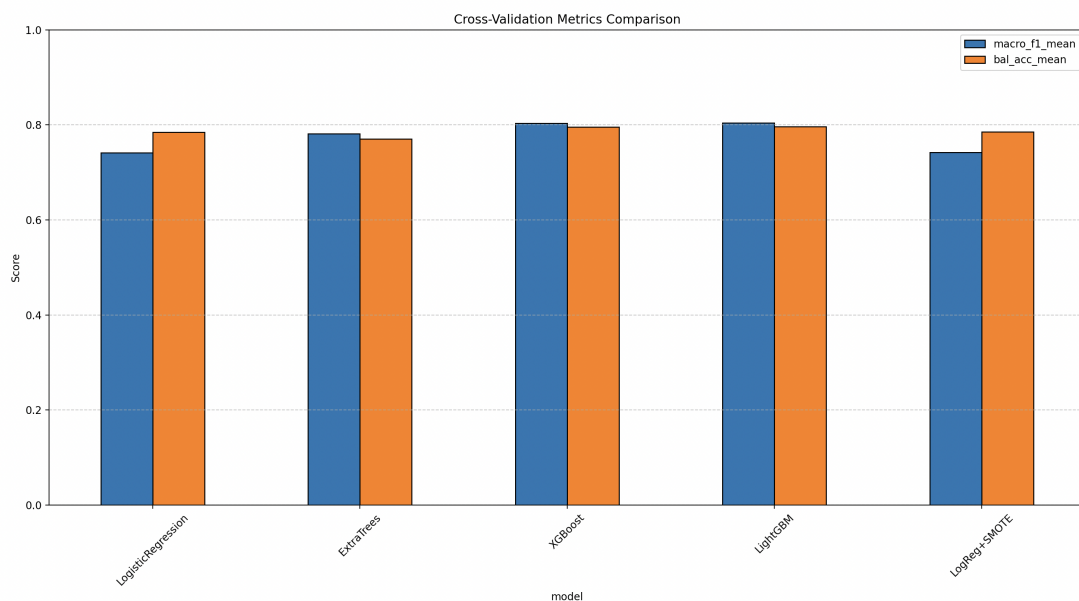


Figure 2: Cross-validation metric comparison across the five candidate models.

The cross-validation results compare the macro F1 score and the balanced accuracy of all the classifiers used in our algorithm, including logistic regression with class weights, Extra Trees, XGBoost, LightGBM, and logistic regression with SMOTE oversampling. It was observed that the tree-based ensemble models (Extra Trees, XGBoost and LightGBM) performed better than the linear models, which suggests that the relationship between the features and accident severity is non-linear and is captured better by models that model hierarchical feature interactions. The best chosen model is LightGBM, as it has the highest macro F1 score, making it the most balanced model for identifying both accident severity classes. Logistic regression with SMOTE shows better performance than the standard logistic regression model, showing that class imbalance is a factor affecting performance. Since Extra Trees has built-in class weights, it gives a strong balanced accuracy score, showing its reliability in handling class imbalance. Overall, the validation results indicate that the gradient boosting models are the most effective ones for this dataset.

3.2 Feature Importance Analysis

The input factors that have the greatest influence on accident severity prediction are highlighted in the feature importance plot for the best model (LightGBM).

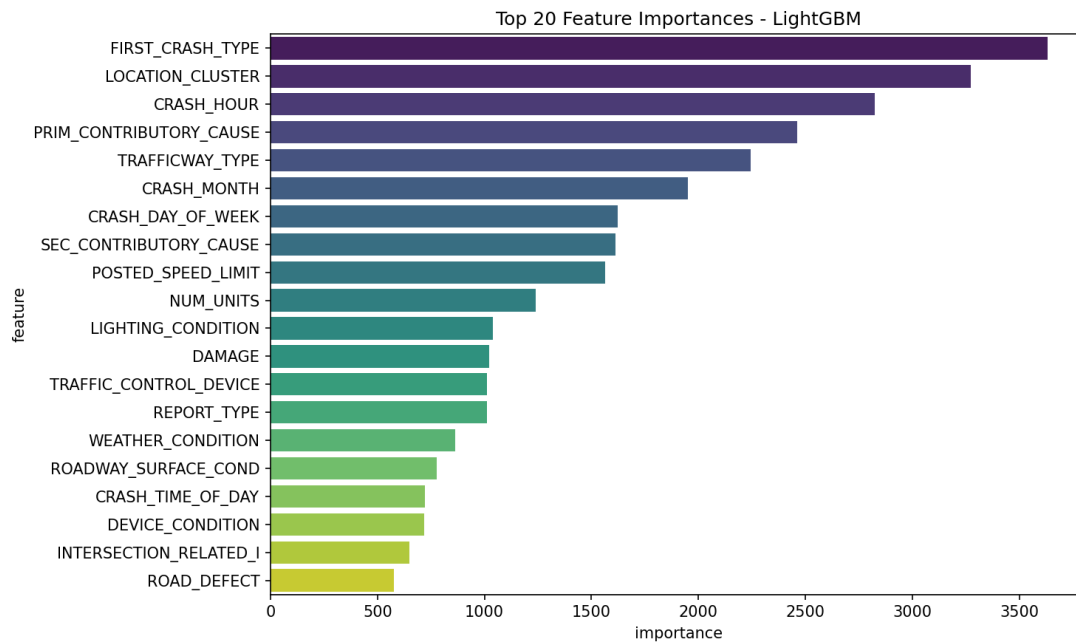


Figure 3: Top 20 feature importances for the LightGBM model.

The model relies largely on speed-related and collision-impact factors, showing that accident dynamics are highly predictive of severity. Features linked to vehicle movement, traffic control presence and road conditions appear among the top predictors, agreeing with real-world accident factors. A number of environmental factors, including weather, lighting and road surface, also rate highly, indicating that external factors affect severe outcomes. Predicting crash severity is multidimensional, requiring a combination of roadway, environmental and driver-behaviour factors, as indicated by the presence of multiple mid-importance elements.

3.3 Confusion Matrix Interpretation

The confusion matrix below shows how well the model differentiates between the two severity classes.

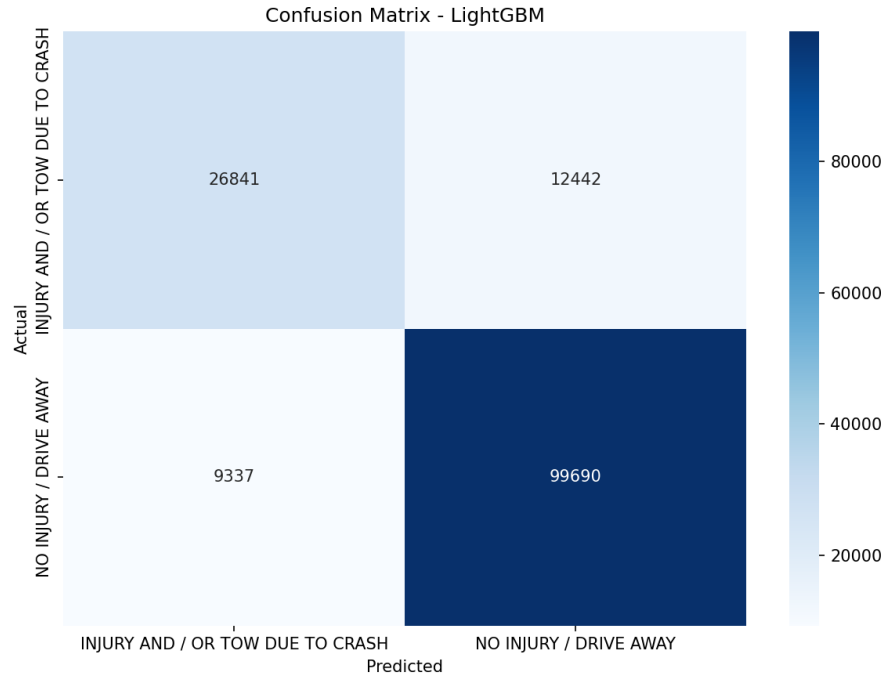


Figure 4: Confusion matrix for the LightGBM model on the held-out test set.

Findings:

1. Most samples in both groups are correctly predicted by the model, indicating strong overall performance.
2. The majority of misclassifications happen in the direction of low severity $\rightarrow$ high severity and vice versa. This is expected, as real-world accident severity commonly lies on a continuum.
3. The comparatively low incidence of false negatives is significant, because under-reporting serious accidents could have detrimental effects on public safety.
4. High diagonal dominance suggests that the classifier generalises well and does not merely memorise the training data.

3.4 Test Performance Metrics

The final evaluation on the held-out test set (`reports/metrics.json`) gives:

Table 2: LightGBM performance on the held-out test set.

Metric	Value
Balanced accuracy	0.7988 (79.88%)
Macro F1 score	0.8065

Even in the presence of class imbalance, the high balanced accuracy indicates that both severity classes are reasonably well predicted. The macro F1 score validates that the model maintains balanced precision and recall rather than favouring the dominant class. The test results are consistent with the cross-validation findings, demonstrating that the model did not overfit and generalises effectively.

From the results it is evident that the most dependable and reliable model for predicting accident severity is LightGBM. Accident severity is controlled by several interacting variables, which makes tree-based models useful. The model demonstrates strong generalisation ability and

performs consistently across the training, validation and test datasets. Feature relevance aligns with real-world crash dynamics, adding interpretability to the final model. This establishes the effectiveness of the selected machine learning approach in predicting accident severity and highlights the potential for applying the model in traffic safety analytics and roadway risk assessment.

3.5 Explainability Methods

To evaluate whether the LightGBM model was making correct and trustworthy predictions, we incorporated Explainable AI (XAI) techniques into the analysis. The goal was to understand how the dataset was being used inside the model and which factors were influencing accident-severity outcomes. By translating the model’s internal mechanics into human-interpretable explanations, XAI allowed us to assess whether predictions aligned with realistic crash-severity patterns and domain expectations.

We added explainability because gradient-boosted tree models like LightGBM, while highly accurate, tend to operate as “black boxes”. Since our task involves predicting road-accident severity, we needed to confirm not just strong statistical performance but also that the model’s decisions were grounded in reasonable, interpretable patterns. Hence XAI methods were used to:

- check whether the model relied on meaningful accident-related factors (e.g. speed, damage, number of units);
- confirm that predictions were logical for individual samples, especially borderline cases;
- translate predictions into actionable insights that could support safety analysis or decision-making.

To support this analysis, two complementary explainability techniques were applied: SHAP and LIME. Both methods are widely used in machine learning because they provide structured, interpretable insights into how models generate predictions. SHAP offers a theoretically grounded framework for quantifying each feature’s contribution, allowing both global patterns and individual decisions to be examined in a consistent manner. LIME, in contrast, focuses on local interpretability by approximating the model’s behaviour around a specific data point, making it useful for validating whether individual predictions are based on meaningful and plausible relationships. Together, these methods help ensure that complex models behave transparently, reveal which inputs drive predictions and allow practitioners to assess whether the model’s reasoning aligns with real-world logic.

3.5.1 SHAP Analysis

SHAP (SHapley Additive exPlanations) was used to examine how each feature contributed to the model’s predictions at both the global and local levels.

Globally, SHAP enabled us to determine which variables consistently exerted the strongest influence on crash-severity outcomes. Features such as `REPORT_TYPE`, `NUM_UNITS`, `DAMAGE` and `POSTED_SPEED_LIMIT` showed the highest overall contributions, confirming that the model relied on meaningful accident-related information rather than noise or artefacts. A positive SHAP value indicated movement toward the Injury class, while a negative value pushed the prediction toward No Injury, with the magnitude reflecting the strength of each effect.

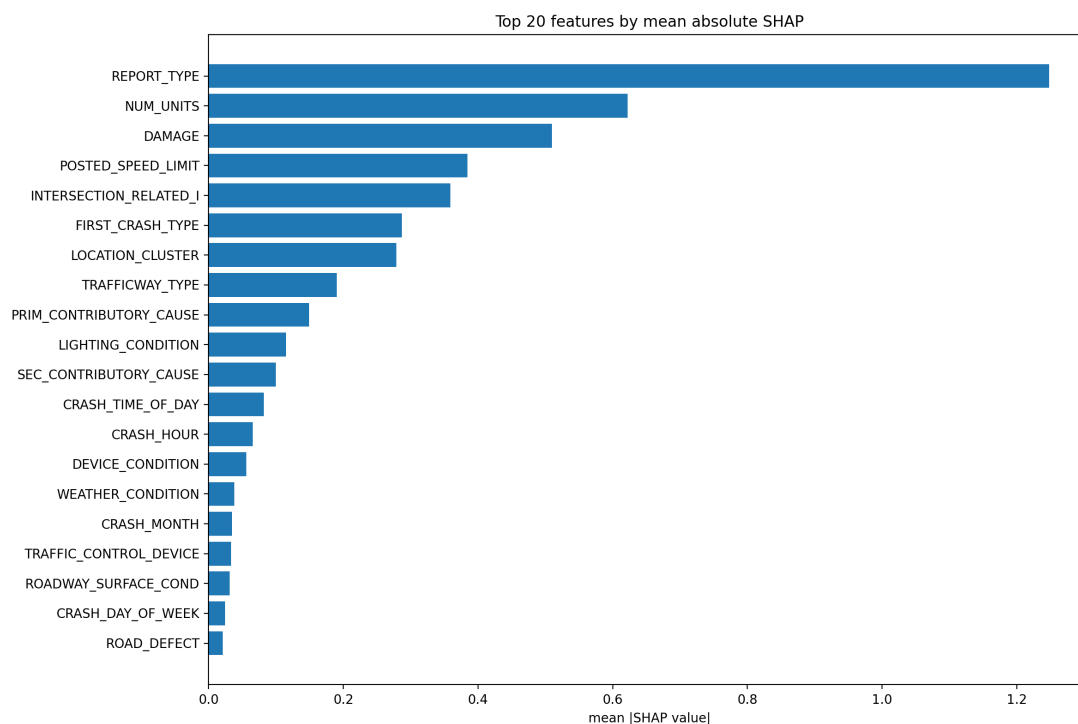


Figure 5: Top 20 features by mean absolute SHAP value.

The results were derived from the average magnitude of SHAP values, and show how strongly each input variable shaped predictions across the entire dataset. **REPORT_TYPE** was the most influential feature, indicating that the type of crash report filed by police was an important signal for distinguishing between injury and non-injury events, suggesting that incidents formally documented by police tend to involve more severe outcomes. Features such as **NUM_UNITS**, **DAMAGE** and **POSTED_SPEED_LIMIT** followed closely, reflecting real-world crash severity outcomes: higher posted speed limits and greater damage levels naturally align with increased injury likelihood, while simple two-unit collisions tend to reduce predicted severity. The fact that the strongest SHAP values were concentrated within this small group of features indicates that the model was focusing on relationships that matter for predicting crash severity, rather than picking up random patterns in the data. Just as importantly, the very low SHAP values for less relevant variables showed that those features had little influence on the predictions.

Furthermore, to see clearly how much each feature contributed to the crash, a percent contribution was calculated for each feature. This was calculated by dividing the mean absolute SHAP value of that feature by the sum of all mean absolute SHAP values across the sampled predictions. This approach converts raw SHAP magnitudes into an interpretable metric that expresses how much of the model’s total decision-making “weight” can be attributed to each variable. For example, **REPORT_TYPE** has a mean absolute SHAP value of 1.25 out of a combined total of 4.63, which corresponds to approximately 26.93% of the total influence. Presenting the results in percentages makes it easier to compare relative importance and identify which features consistently contributed the most to the model’s understanding of injury likelihood. This metric complements traditional feature importance from LightGBM by grounding the interpretation in the actual direction and magnitude of contribution toward the severity prediction.

Table 3: Top 20 features by percent contribution to the model’s total SHAP magnitude, with suggested preventive measures.

Feature	Percent	Suggested preventive measure
REPORT_TYPE	26.93%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
NUM_UNITS	13.42%	Improve intersection control and sightlines; redesign merges/diverges.
DAMAGE	11.01%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
POSTED_SPEED_LIMIT	8.30%	Lower speed limits where needed; speed cameras and traffic calming.
INTERSECTION_RELATED_I	7.76%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
FIRST_CRASH_TYPE	6.19%	Target high-risk collision types with engineering (roundabouts, protected turns), education and enforcement.
LOCATION_CLUSTER	6.02%	Install engineering countermeasures (speed cushions, signage), hotspot redesigns and increased enforcement.
TRAFFICWAY_TYPE	4.11%	Road design changes: medians, separated lanes and speed management.
PRIM_CONTRIBUTORY_CAUSE	3.23%	Behavioural interventions: distracted-driving campaigns, DUI checkpoints, enforcement for speeding.
LIGHTING_CONDITION	2.49%	Upgrade street lighting and reflective signage.
SEC_CONTRIBUTORY_CAUSE	2.16%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
CRASH_TIME_OF_DAY	1.78%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
CRASH_HOUR	1.42%	Focus lighting, policing and awareness campaigns at high-risk hours.
DEVICE_CONDITION	1.21%	Maintain traffic control devices and signals.
WEATHER_CONDITION	0.83%	Weather-responsive signage and maintenance in seasonal risk months.
CRASH_MONTH	0.75%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
TRAFFIC_CONTROL_DEVICE	0.72%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
ROADWAY_SURFACE_COND	0.68%	Improve pavement maintenance and anti-skid surfacing.
CRASH_DAY_OF_WEEK	0.54%	Investigate and apply targeted engineering, enforcement or education depending on feature context.
ROAD_DEFECT	0.46%	Promptly repair defects and inspect high-risk sections frequently.

Local interpretability using SHAP. SHAP was also used to analyse specific crash samples to understand the exact feature interactions that led to individual prediction outcomes. Local SHAP explanations provide a *force-plot representation* where features pushing the prediction toward the Injury class appear in red, and those pushing it toward the No Injury class appear in blue. The base value represents the model’s expected prediction across the entire dataset, and the final output value results from adding or subtracting the feature contributions for the selected sample.



Figure 6: SHAP force plot for the test sample at index 10.

For the sample positioned at index 10 in the test set, SHAP revealed that the model predicted a low likelihood of injury because several strong negative (blue) factors dominated the decision. These included a low POSTED_SPEED_LIMIT, a minimal DAMAGE level, a benign crash REPORT_TYPE and a simple two-unit involvement, all of which historically correlate with non-injury outcomes. Although a few contextual variables such as lighting conditions or trafficway type pushed the prediction slightly toward the Injury side, their influence was comparatively weak and insufficient to override the strong indicators of a mild crash. This level of granularity helped validate that the model's prediction for this specific incident was grounded in a realistic and contextually coherent explanation.

3.5.2 LIME Analysis

LIME (Local Interpretable Model-agnostic Explanations) was used as a complementary technique to validate the model's reasoning at the individual-case level. Unlike SHAP, which explains model behaviour across the dataset, LIME constructs a simplified local approximation of the model around a specific prediction. This allowed us to check whether small, realistic changes in feature values would meaningfully alter the predicted severity, and whether those changes aligned with expectations.

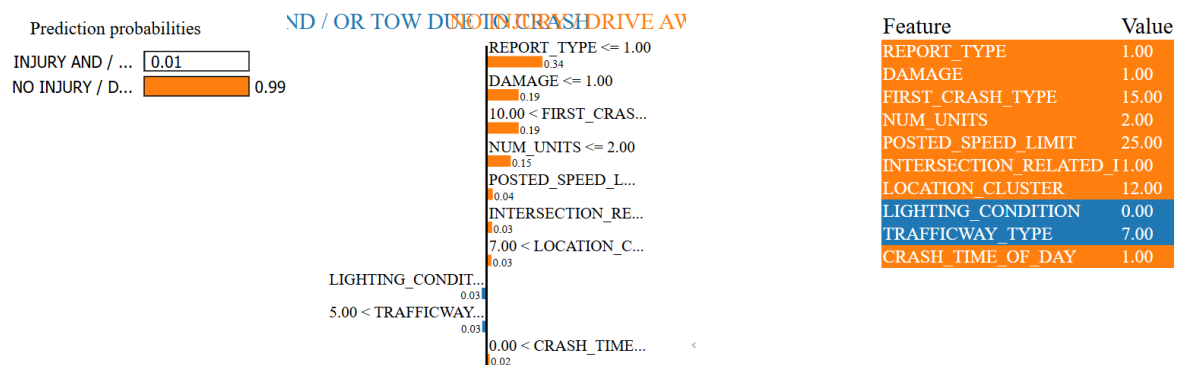


Figure 7: LIME explanation for the same test sample at index 10.

LIME highlighted a low-severity prediction, meaning the No Injury class. Consistent with the SHAP results, LIME identified a low POSTED_SPEED_LIMIT, minimal DAMAGE, a typical crash location and a non-critical REPORT_TYPE as strong indicators of a mild crash. While certain conditions such as LIGHTING_CONDITION and TRAFFICWAY_TYPE were shown to moderately increase the predicted severity, LIME's explanation made clear that these factors were outweighed by multiple dominant features signalling a low-risk event. The convergence between LIME and SHAP provided strong evidence that the model's prediction for this sample was stable, interpretable and aligned with domain knowledge.

3.5.3 Relevance of Explainability to the Overall Analysis

Using both SHAP and LIME generated a comprehensive, multi-level understanding of why the LightGBM model behaved the way it did. Globally, the explainability methods demonstrated that the model relied heavily on logically meaningful features, such as report type, speed limits,

number of units involved and damage severity, when predicting crash outcomes. This confirms that the model was not driven by noise, spurious correlations or dataset artefacts.

At the local level, SHAP and LIME validated the correctness of predictions for specific crash cases by exposing the precise variable interactions responsible for the outcome. When both methods agreed on the reasoning behind a prediction, it strengthened confidence that the model's internal logic was consistent and trustworthy. This approach was essential for ensuring that the model could not only achieve strong predictive accuracy but also provide insights suitable for informed decision-making in road-safety analysis.

Explainability hence helped show that the LightGBM model was a transparent analytical tool capable of supporting evidence-based discussions about accident severity. The alignment between real-world crash patterns and the model's inferred logic shows the reliability of its predictions.

3.6 Geospatial Analysis

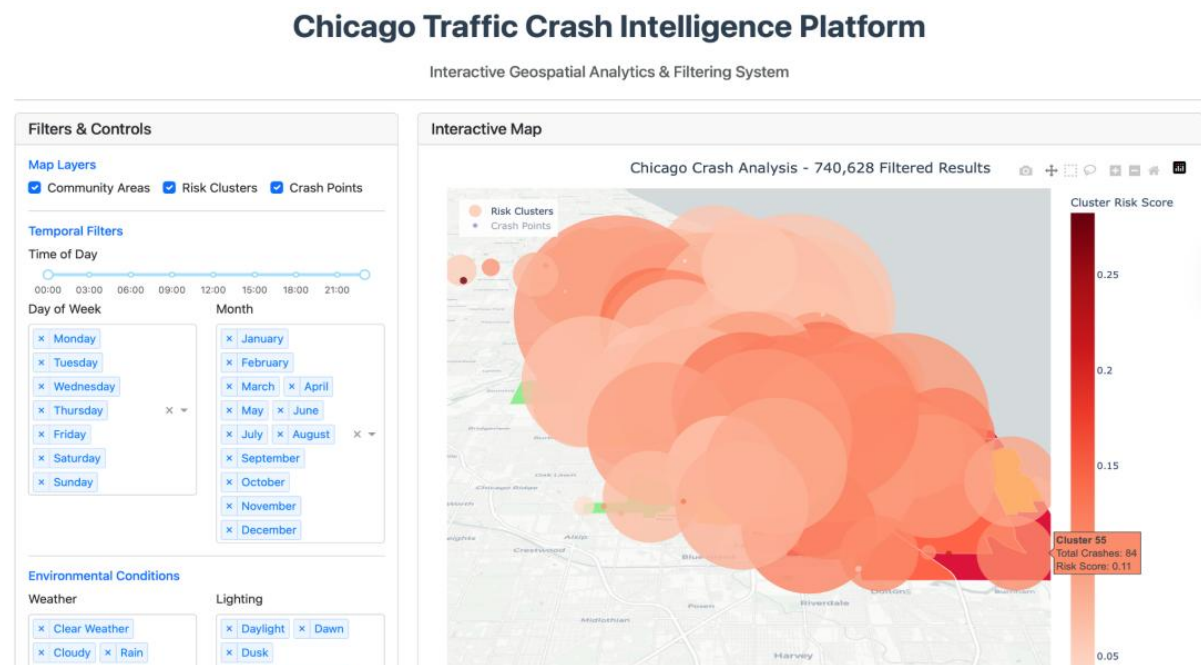


Figure 8: Representation of individual clusters and their corresponding risk score.

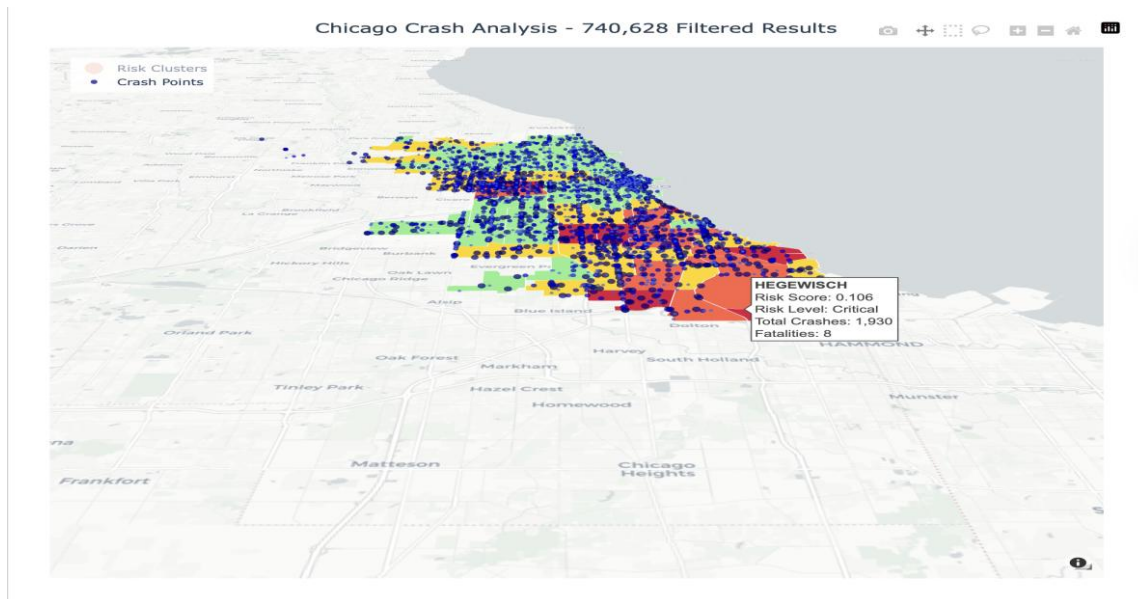


Figure 9: Individual crash points spread out across Chicago.



Figure 10: Dynamic modification of parameters leading to a change in the mean risk score. Restricting the view to fatal and injury crashes raises the mean risk score from the 0.15 baseline to 1.57, an increase of 951.5%.